



Introduction to Quantum Computing

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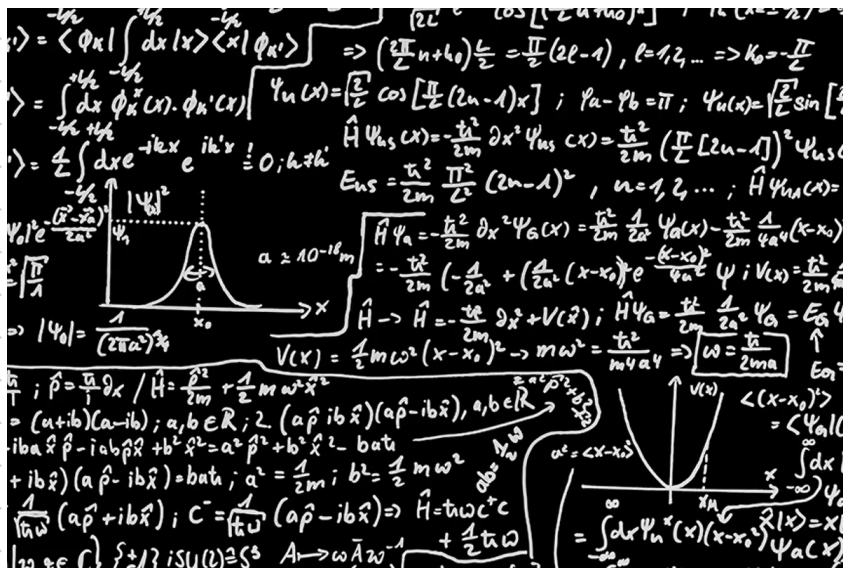
Talk by Anyaman Kolhe
@ Qiskit Fall Fest 2025,
IIT Hyderabad.

What to expect :

Just the bare minimum math to get you started.

And enough to get you interested!

Tuff math?



<https://isaactengchithang.wordpress.com/2024/06/15/quantum-mechanics-an-introduction/>

(Yes, but we can start small)

Prerequisites :

- * Complex numbers
- * Probability
- * Matrix Multiplication

$$z = a + ib \in \mathbb{C} \quad a, b \in \mathbb{R}, \quad i = \sqrt{-1}$$

$$\sum_i P_i(E_i) = 1, \quad 0 \leq P_i(E_i) \leq 1 \quad \forall_i$$

$$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{2 \times 3} \begin{bmatrix} \vdots \\ \vdots \end{bmatrix}_{3 \times 1} = \begin{bmatrix} \vdots \end{bmatrix}_{2 \times 1}$$

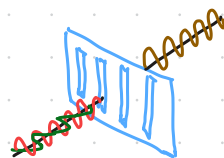
Quantum !!

Classical Bits  Eg: implemented through wires.

But we abstract this out with binary digits : 0, 1
[bits]

E.g. $\{0, 1\}^2 = \{00, 01, 10, 11\}$

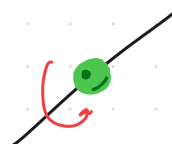
Quantum Bits : Observed as



Horizontal $|0\rangle$
or
Vertical $|1\rangle$
Polarization of light

figures for representative purposes only!

Or :



Spin $+\frac{1}{2}$ $|0\rangle$
Spin $-\frac{1}{2}$ $|1\rangle$
of an electron

Think of a qubit $|\psi\rangle$ as an abstraction the way we use 0, 1 for classical information.

Dirac's Bra-Ket Notation

$$|a\rangle = \text{"Ket } a\text{"}$$

$$\text{e.g.: } |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}_{2 \times 1}$$

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}_{2 \times 1}$$

For a Qubit

$$\langle b| = \text{"Bra } b\text{"}$$

$$= (|b\rangle)^\dagger$$

$$= (|b\rangle^*)^T$$

Conjugate
transpose

$$\text{e.g.: } \begin{bmatrix} 1 & 1+i \\ 3 & 4-i \end{bmatrix}^\dagger = \begin{bmatrix} 1 & 3 \\ 1-i & 4+i \end{bmatrix}$$

If your vector was $d \times 1$, e.g.: $|0\rangle = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{d \times 1}$

then we call it a qubit

We stick to $d=2$ (qubits) for now.

When popular explanations say that a qubit can be "both 0 & 1" at the same time, what they really mean is that an arbitrary qubit can be written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where $\alpha, \beta \in \mathbb{C}$

$$= \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

Measurement

(along a certain basis, say $\{|0\rangle, |1\rangle\}$)

e.g. Physically, it can be measuring an electron's spin

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

↳ Probability amplitudes

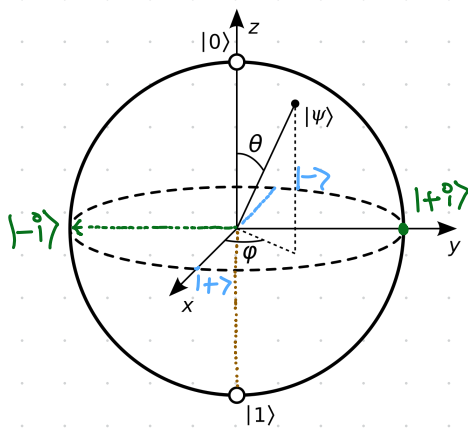
$$\text{Then, } P_2[|\psi\rangle \rightarrow |0\rangle] = |\alpha|^2$$

$$P_2[|\psi\rangle \rightarrow |1\rangle] = |\beta|^2$$

From here we see that

$$|\alpha|^2 + |\beta|^2 = 1 \quad (\text{Normalization})$$

Bloch Sphere



https://en.wikipedia.org/wiki/Bloch_sphere

Pole States

$$Z^+ \text{ pole} : |0\rangle$$

$$Z^- \text{ pole} : |1\rangle$$

$$X^+ \text{ pole} : |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$X^- \text{ pole} : |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$Y^+ \text{ pole} : |+\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$$

$$Y^- \text{ pole} : |-\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

HW: If $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

s.t. $|\alpha|^2 + |\beta|^2 = 1$ then we can

represent the state as $|\psi\rangle = \cos\theta/2 |0\rangle + e^{i\phi} \sin\theta/2 |1\rangle$

where θ, ϕ are angles as shown in figure.

Note how if we have $|\psi\rangle = |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$ then

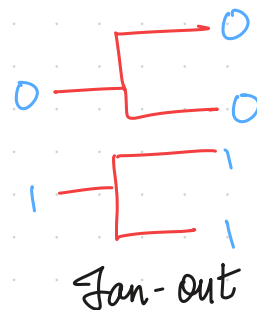
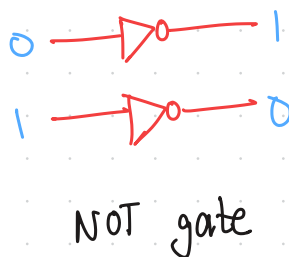
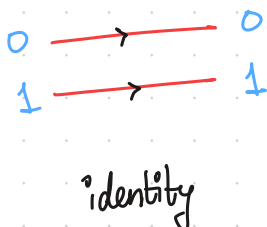
$$\mathbb{P}_2 \left[|+\rangle \rightarrow |0\rangle \right] = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad \text{and}$$

$$\mathbb{P}_2 \left[|+\rangle \rightarrow |1\rangle \right] = \frac{1}{2}$$

$\Rightarrow$ Can generate randomness via measurement

Quantum Logic Gates

Manipulating Classical Bits



Not possible with general $|\psi\rangle$ in Quantum!
See: No Cloning Theorem

You can apply a quantum logic gate on one or more qubits.

Quantum Gates are given by unitary matrices $U^\dagger = U^{-1}$

$$\Rightarrow U^\dagger U = U U^\dagger = \mathbb{1}$$

Think : Why should they be unitary?

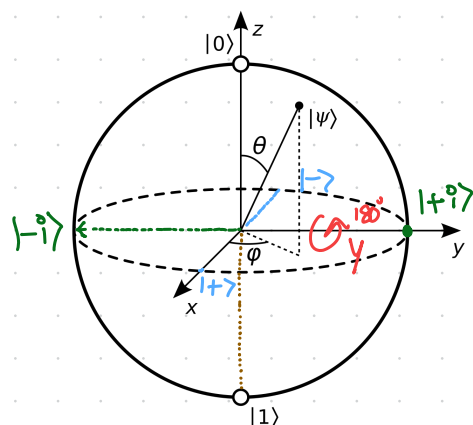
Hint : Recall normalization

Pauli Gates

$$\text{---} \boxed{X} \text{---} \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\text{---} \boxed{Y} \text{---} \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$\text{---} \boxed{Z} \text{---} \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



https://en.wikipedia.org/wiki/Bloch_sphere

How to implement gates

$$\text{Eg. } X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$X|0\rangle = |1\rangle$$

$$X|1\rangle = |0\rangle$$

$$X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

For an arbitrary qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

$$X|\psi\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix} = \beta|1\rangle + \alpha|0\rangle$$

Since we know $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$

$$\begin{aligned} \text{we can directly say } X|\psi\rangle &= X(\alpha|0\rangle + \beta|1\rangle) \\ &= \alpha X|0\rangle + \beta X|1\rangle \\ &= \alpha|1\rangle + \beta|0\rangle \end{aligned}$$

The Hadamard Gate

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$



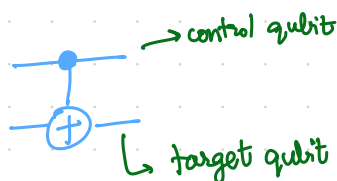
$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle$$

Similarly $H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$

verify this!

Verify that H is unitary!

The Controlled NOT Gate



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Our first multi-qubit gate.

CNOT over 2 bits:



Representing Multi Qubit States

(via tensor products)

$$|0\rangle \otimes |0\rangle = |00\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Tensor product Eg.

Say $A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$

$$B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Then, $A \otimes B = \begin{bmatrix} a_1 B & a_2 B \\ a_3 B & a_4 B \end{bmatrix}$

$$= \begin{bmatrix} a_1 b_1 & a_2 b_1 \\ a_1 b_2 & a_2 b_2 \\ a_3 b_1 & a_4 b_1 \\ a_3 b_2 & a_4 b_2 \end{bmatrix}$$

Action of CNOT gate on qubits:

$$\text{CNOT}(|0\rangle_c \otimes |0\rangle_t) = |0\rangle_c \otimes |0\rangle_t$$

$$\text{CNOT}(|1\rangle_c \otimes |0\rangle_t) = |1\rangle_c \otimes |1\rangle_t$$

$$\text{CNOT}(|0\rangle_c \otimes |1\rangle_t) = |0\rangle_c \otimes |1\rangle_t$$

$$\text{CNOT}(|1\rangle_c \otimes |1\rangle_t) = |1\rangle_c \otimes |0\rangle_t$$

Say $|\psi\rangle$ was a 2 qubit state:

$$|\psi\rangle = \alpha_1 |0\rangle \otimes |0\rangle + \alpha_2 |0\rangle \otimes |1\rangle + \alpha_3 |1\rangle \otimes |0\rangle + \alpha_4 |1\rangle \otimes |1\rangle$$

Then the action of CNOT on $|\psi\rangle$ gives

$$\text{CNOT } |\psi\rangle = \alpha_1 |0\rangle \otimes |0\rangle + \alpha_2 |0\rangle \otimes |1\rangle + \alpha_3 |1\rangle \otimes |1\rangle + \alpha_4 |1\rangle \otimes |0\rangle$$

Entanglement

Not all multi-qubit states can be represented as a tensor product of single qubit states. Such states are called "Entangled"

$$\text{Say } |\psi\rangle = \frac{1}{\sqrt{2}} |0\rangle \otimes |0\rangle + \frac{1}{\sqrt{2}} |0\rangle \otimes |1\rangle$$

$$= |0\rangle \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right)$$

$$= |0\rangle \otimes |+\rangle$$

$\therefore$ Not Entangled!

$$\text{But if } |\psi\rangle = \frac{1}{\sqrt{2}} |0\rangle \otimes |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \otimes |1\rangle$$

Then, say $|\psi\rangle$ can be written as $|\psi_1\rangle \otimes |\psi_2\rangle$

$$(\alpha_1 |0\rangle + \beta_1 |1\rangle) \otimes (\alpha_2 |0\rangle + \beta_2 |1\rangle)$$

Then,

$$\begin{aligned} (\alpha_1 |0\rangle + \beta_1 |1\rangle) \otimes (\alpha_2 |0\rangle + \beta_2 |1\rangle) &= \alpha_1 \alpha_2 |00\rangle + \alpha_1 \beta_2 |01\rangle + \alpha_2 \beta_1 |10\rangle + \beta_1 \beta_2 |11\rangle \\ &= \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle \end{aligned}$$

$$\Rightarrow \alpha_1 \beta_2 = 0 \Rightarrow \text{either } \alpha_1 = 0 \text{ or } \beta_2 = 0 \quad (\text{or both})$$

But, if $\alpha_1 = 0$ then $\alpha_1 \alpha_2 = 0 \neq \frac{1}{\sqrt{2}}$ and if $\beta_2 = 0$ then $\beta_1 \beta_2 = 0 \neq \frac{1}{\sqrt{2}}$

$\therefore$ Such $|\psi_1\rangle, |\psi_2\rangle$ do not exist $\therefore \frac{|00\rangle + |11\rangle}{\sqrt{2}}$ is entangled!

What entanglement means physically:

If you had $|\phi\rangle = \frac{1}{\sqrt{2}} (|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B)$

and if Alice with register A measured her qubit in $\{|0\rangle, |1\rangle\}$ and got $|0\rangle_A$, then Bob's state will definitively collapse to $|0\rangle_B$

———— x ————